Projet crypto-bot

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Plan

- Objetif
- Docker compose
- Fast API
- Ingestion
- Machine Learning
- Base de données
- Pipelines
- Tests unitaires
- CI/CD
- Dashboard
- Conclusion / Améliorations

Objectif

Catalogue → **Pipeline** → **Dashboard**

- *Sélection de monnaies*

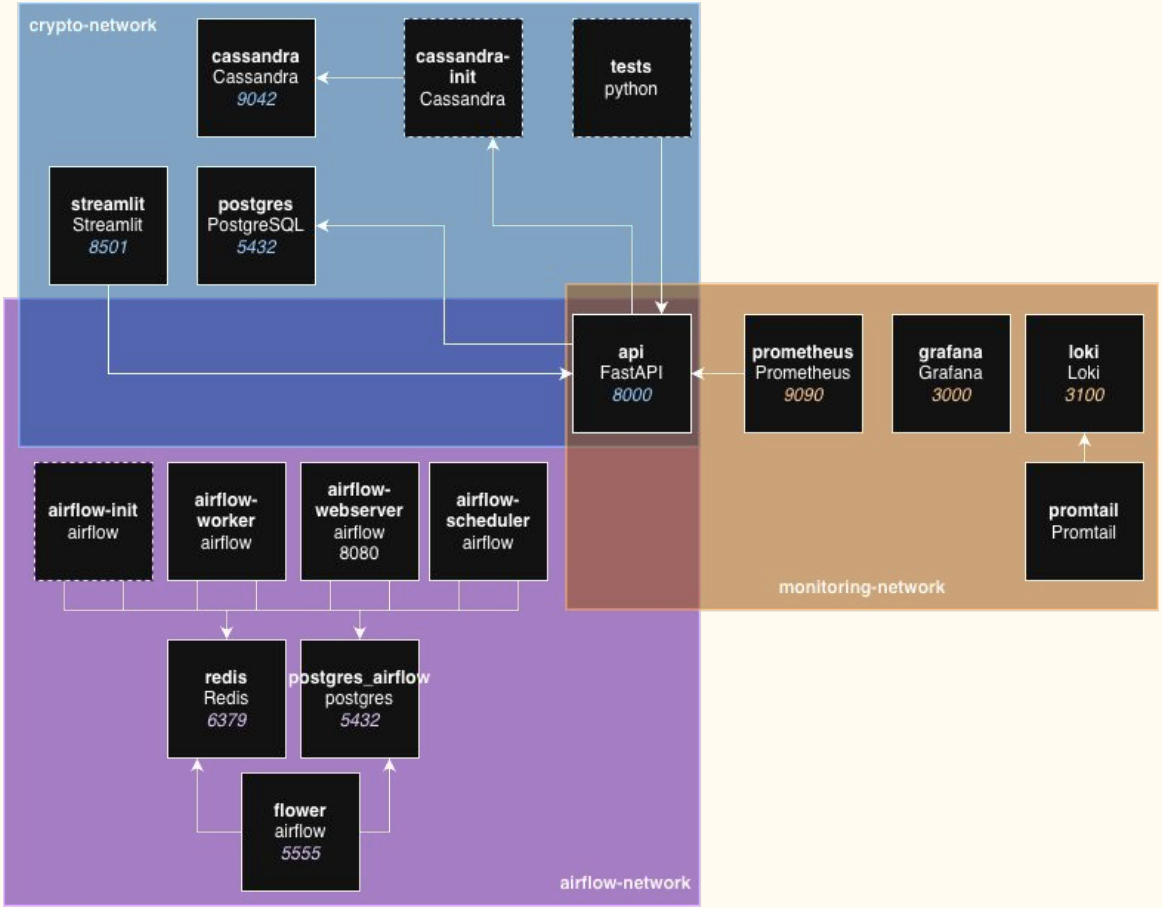
- *Affichage cours / prédiction*

Structure du projet

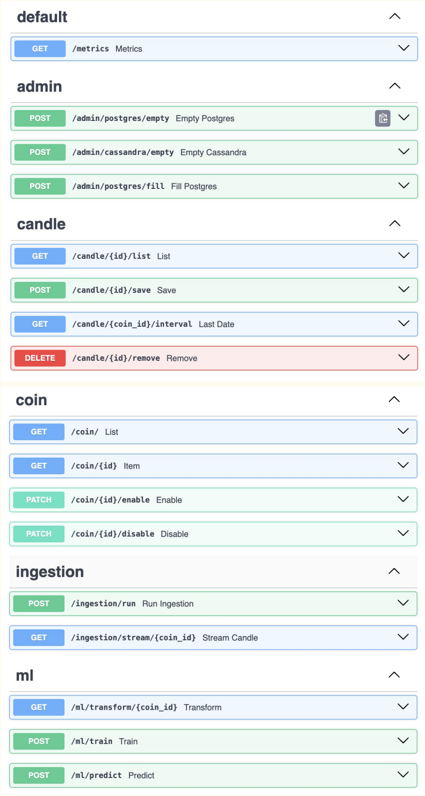
Docker compose

```yaml
include:
  - docker-compose.airflow.yml
  - docker-compose.cassandra.yml
```

```yaml
networks:
  airflow-network:
  crypto-network:
  monitoring-network:
```



FastAPI



- Initialisation de Cassandra

- Initialisation de FastAPI
- Middleware d'autorisation des services externs

- Déclaration des routes

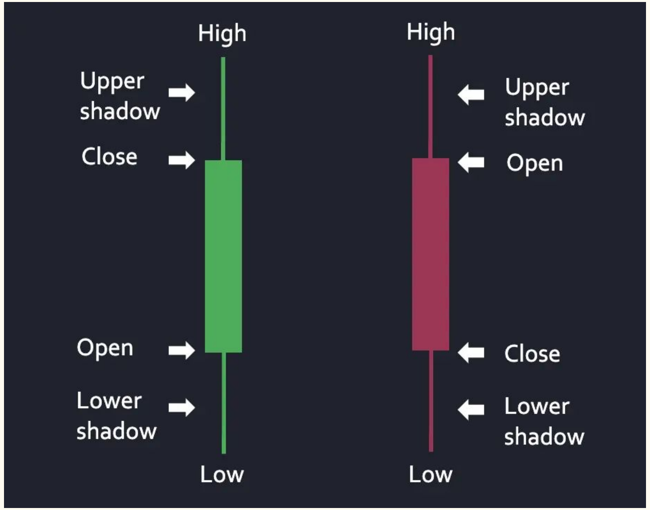
```python
from cassandra.cluster import Cluster
from contextlib import asynccontextmanager
from fastapi import FastAPI
from fastapi.middleware.cors import CORSMiddleware
from prometheus_fastapi_instrumentator import Instrumentator

from routes import admin as admin_router
from routes import candle as candle_router
from routes import coin as coin_router
from routes import ingestion as ingestion_router
from routes import ml as ml_router

@asynccontextmanager
async def lifespan(app: FastAPI):
    CLUSTER_IPS = ['cassandra']
    keyspace = "crypto_bot"
    cluster = Cluster(CLUSTER_IPS, compression=True)
    session = cluster.connect(keyspace=keyspace)
    app.state.cassandra_session = session
    yield
    cluster.shutdown()

app = FastAPI(title="API FastAPI du projet crypto-bot", lifespan=lifespan)
Instrumentator().instrument(app).expose(app)

app.add_middleware(
    CORSMiddleware,
    allow_origins=["*"],
    allow_credentials=True,
    allow_methods=["*"],
    allow_headers=["*"],
)

app.include_router(admin_router.router, prefix="/admin", tags=["admin"])
app.include_router(candle_router.router, prefix="/candle", tags=["candle"])
app.include_router(coin_router.router, prefix="/coin", tags=["coin"])
app.include_router(ingestion_router.router, prefix="/ingestion", tags=["ingestion"])
app.include_router(ml_router.router, prefix="/ml", tags=["ml"])
```

Ingestion

Format bougie (k-line)

- open
- high
- low
- close
- volume

(ohlcv)



CCXT

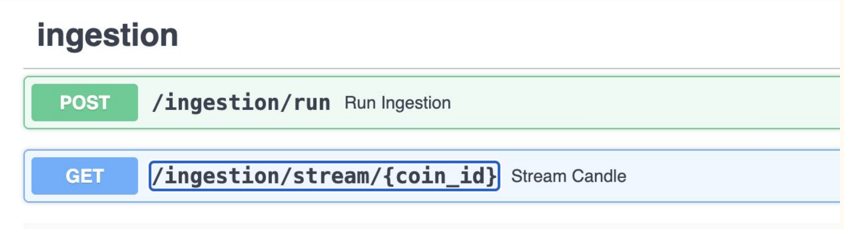
Utilisation

- Classique pour la récupération du passé
- Websocket pour la récupération du présent

Réflexivité

```
exchange_id = "binance"
exchange_class = getattr(ccxt, exchange_id, None)
```

Routes



CoinGecko

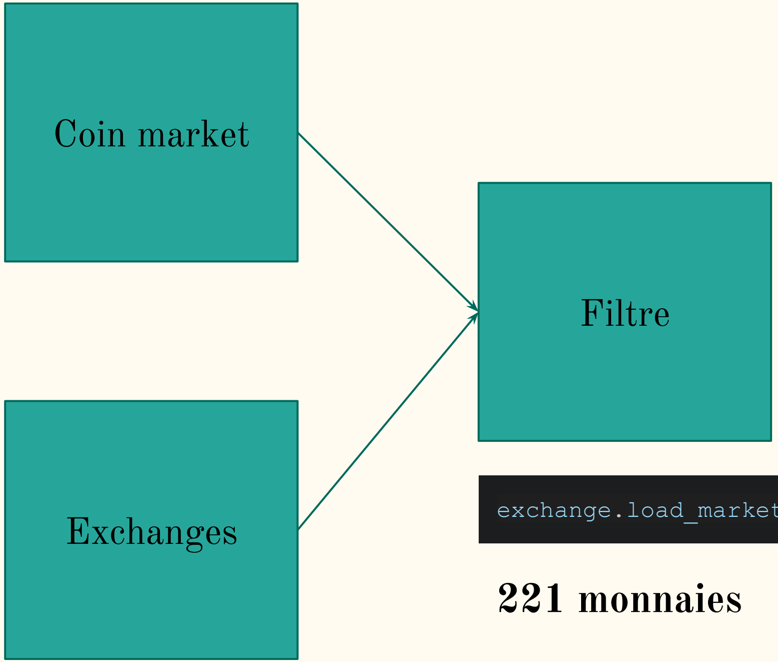
Contruction du catalogue :

```
coin_markets = client.coins.markets.get(
        vs_currency='eur',
        per_page=length,
        page=1
    )
```

250 monnaies

```
exchanges = ccxt.exchanges
```

(CCXT)




```
exchange.load_markets().values()
```

221 monnaies

Machine Learning

Contexte

Difficultées rencontrées

- Série temporelle
- Micro-variations (bruit)
- Choix de la cible de prédiction

Traitement

- Traitement de la cible
    - Lissage (Moyenne)
- Création de features
    - Informations du passé (10 points)
- Echantillonage des données
    - Choix du pas de récupération des données (**1h**)
- Choix de la cible (prédiction à **j+1**)

Choix du modèle

Classification

- **0** : baisse / pas de variation : vendre
- **1** : hausse : achat

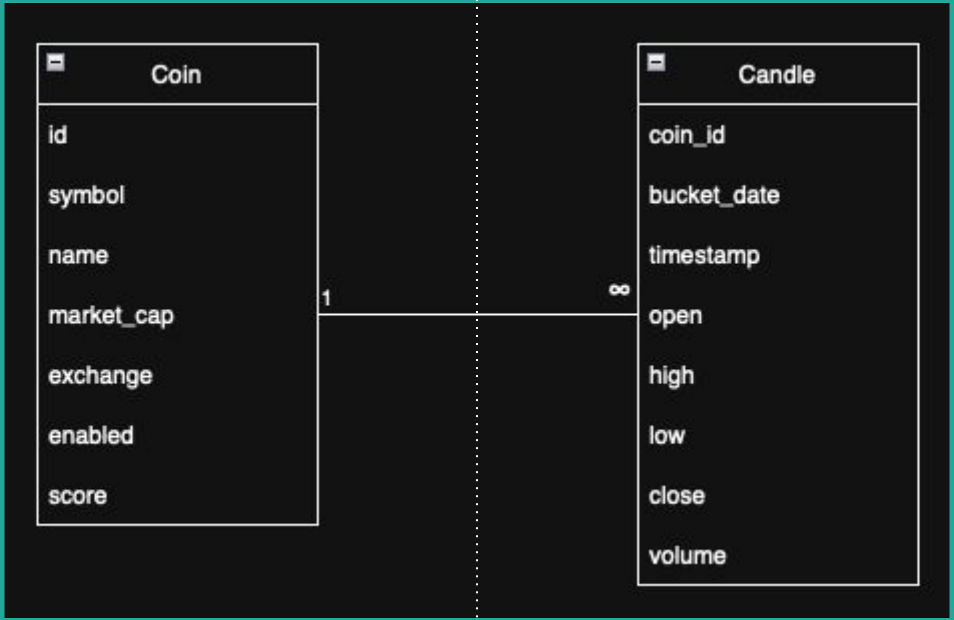
Modèle	Score
RandomForestClassifier	**0.59**
GradientBoostingClassifier	0.52
LogisticRegression	0.55
DecisionTreeClassifier	0.54

Score

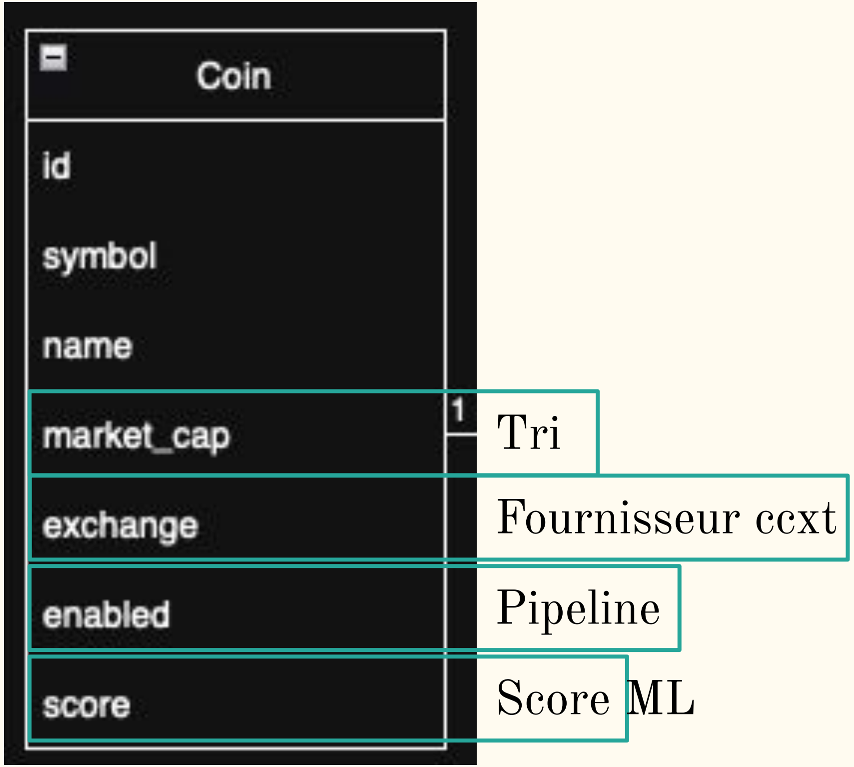
- Faux négatif pas important dans notre contexte
- **Faux positif** à minimiser => **précision**

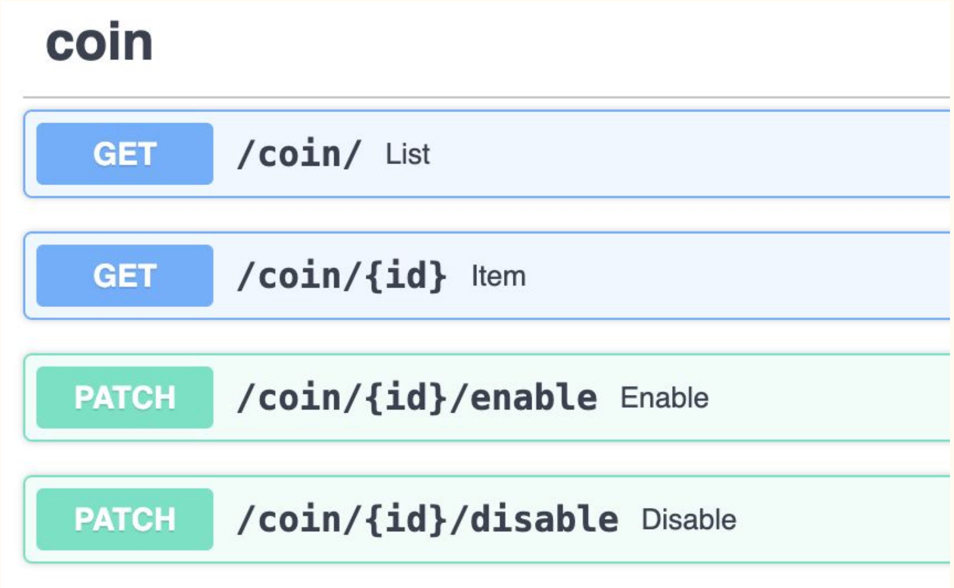
Base(s) de données

PostgreSQL Cassandra



PostgreSQL


Coin
id
symbol
name
market_cap
exchange
enabled
score
1
Tri
Fournisseur ccxt
Pipeline
Score ML


coin
GET /coin/ List
GET /coin/{id} Item
PATCH /coin/{id}/enable Enable
PATCH /coin/{id}/disable Disable

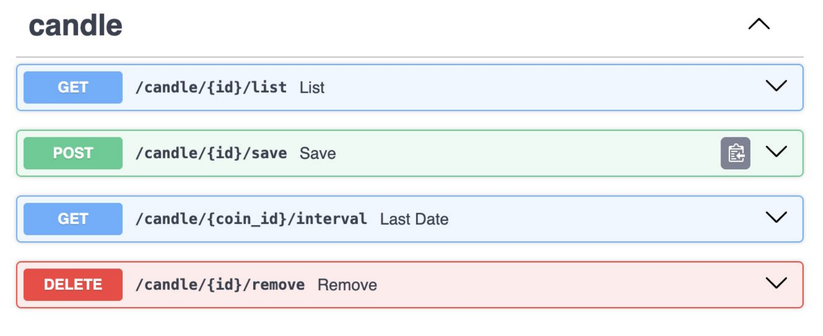
Cassandra

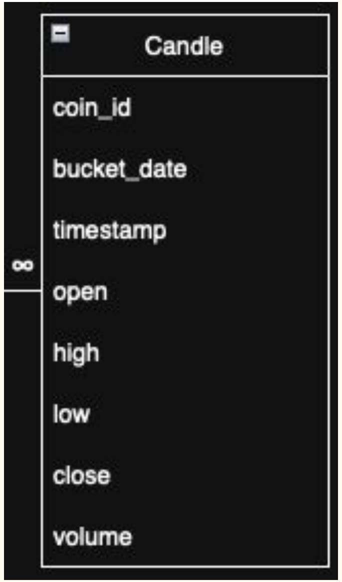
Recommendations :

- Moins de 100 000 lignes par partions
- Partition de 100 Mo Max
- Moins de 10 Mo par ligne

Partitionement (pas de 1 heure) :

- Clé de partitionnement : **crypto_id** + **bucket_date**
- Cle de clustering : **timestamp**
- Taille par ligne : 16 (coin_id)+10 (bucket_date)+64 (timestamp)+8 (open)+8 (high)+8 (low)+8 (close)+8 (volume)=**130 octets**
- **8784 lignes** (366*24)
- **1,1 Mo par partition** (8784 * 130)





Pipeline Airflow

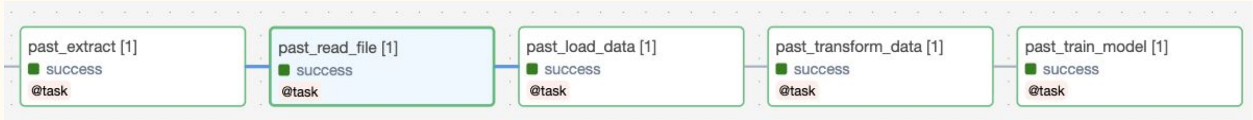
Airflow

Factorisation des pipelines :

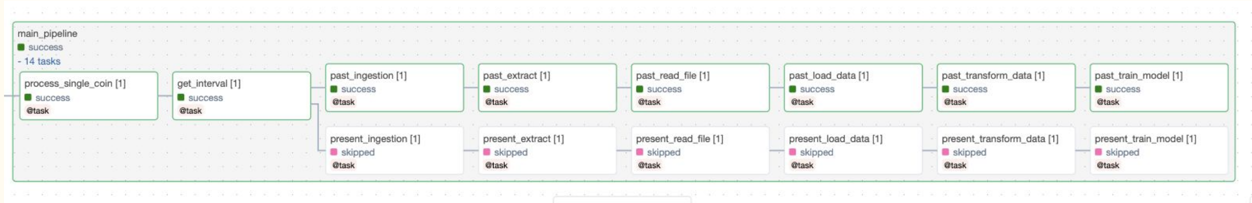
- Dynamic Task Mapping
- Task group
- Duplication de tâches
- Appel manuels

Task groups :

ELT :

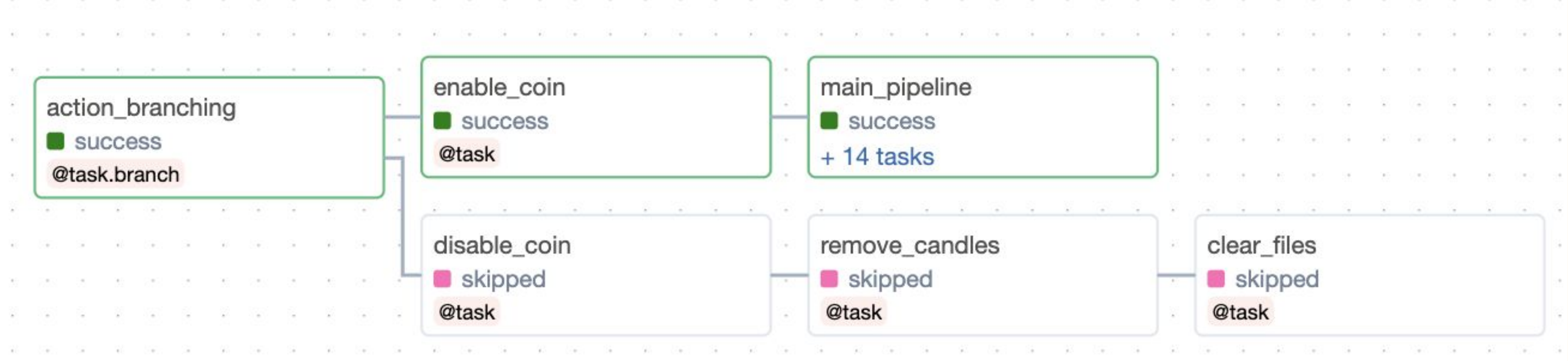


Pipeline principal :



Airflow

Pipeline manuel :



Airflow

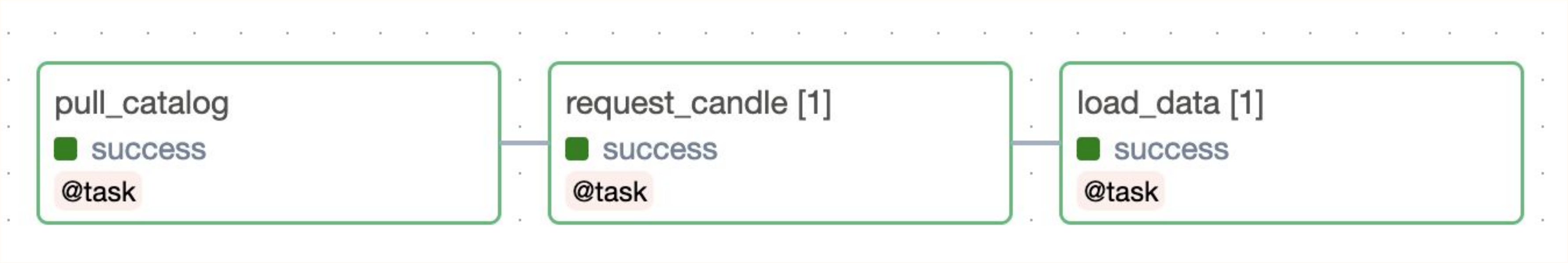
Ingestion :

(Chaque mois)

Airflow

Streaming :

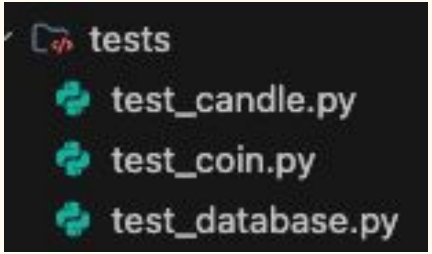
(Chaque heure)



Tests unitaires

Pytest pour FastAPI

```yaml
tests:
  image: python:3.11-slim
  container_name: tests
  profiles:
    - test
  volumes:
    - ./tests:/app
  working_dir: /app
  environment:
    - PYTHONPATH=/app
  depends_on:
    api:
      condition: service_healthy
  entrypoint: >
   sh -c " pip install httpx && pip
install pytest && pytest -v "
  networks:
    crypto-network:
```

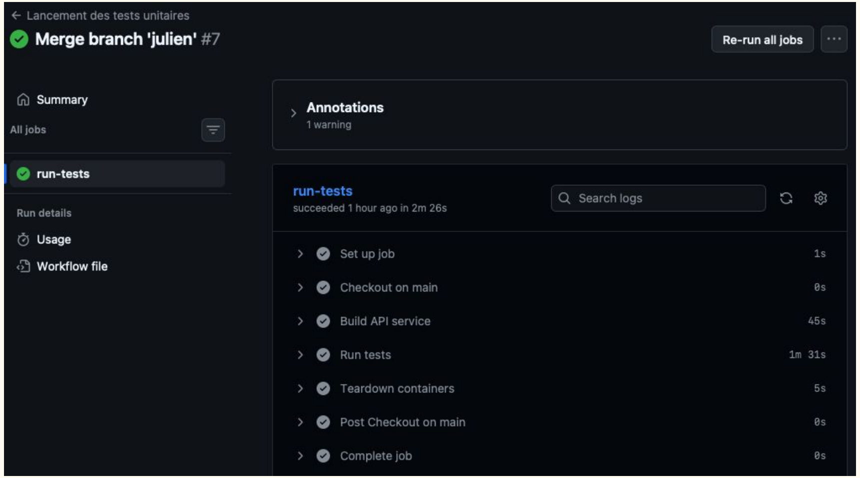



```python
import httpx

def test_root():
   url = "http://api:8000/"
   response = httpx.get(url)
   assert response.status_code == 404
```

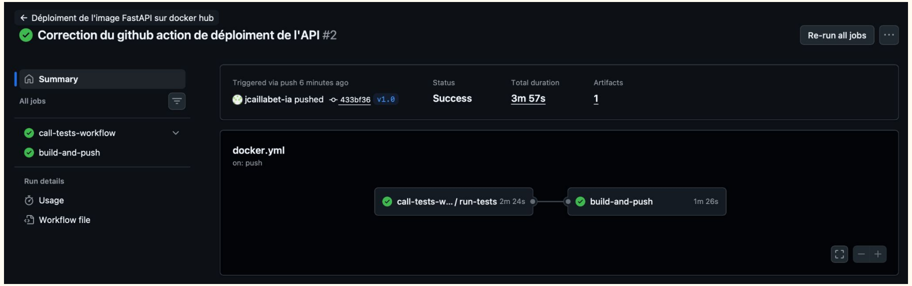
Pipelines CI/CD

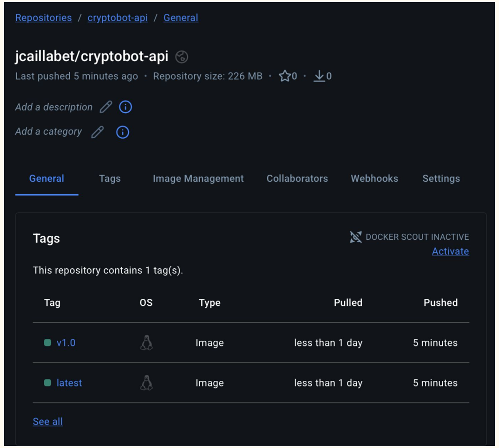
Github actions

Test unitaires



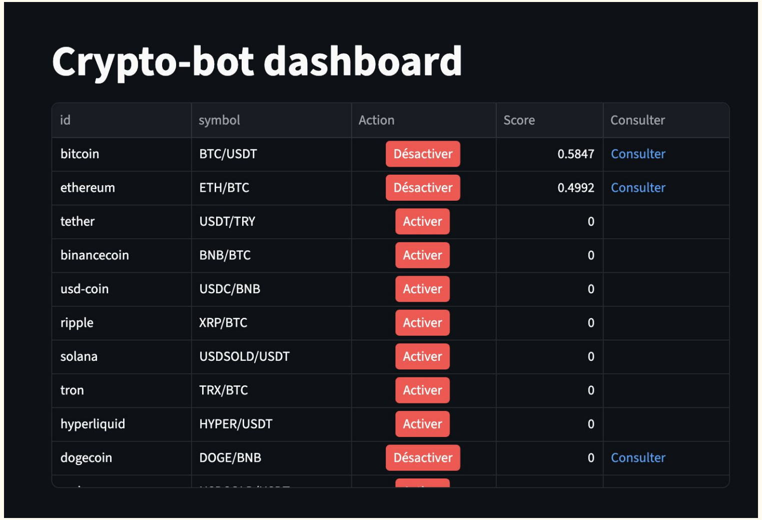
Build image





Dashboard

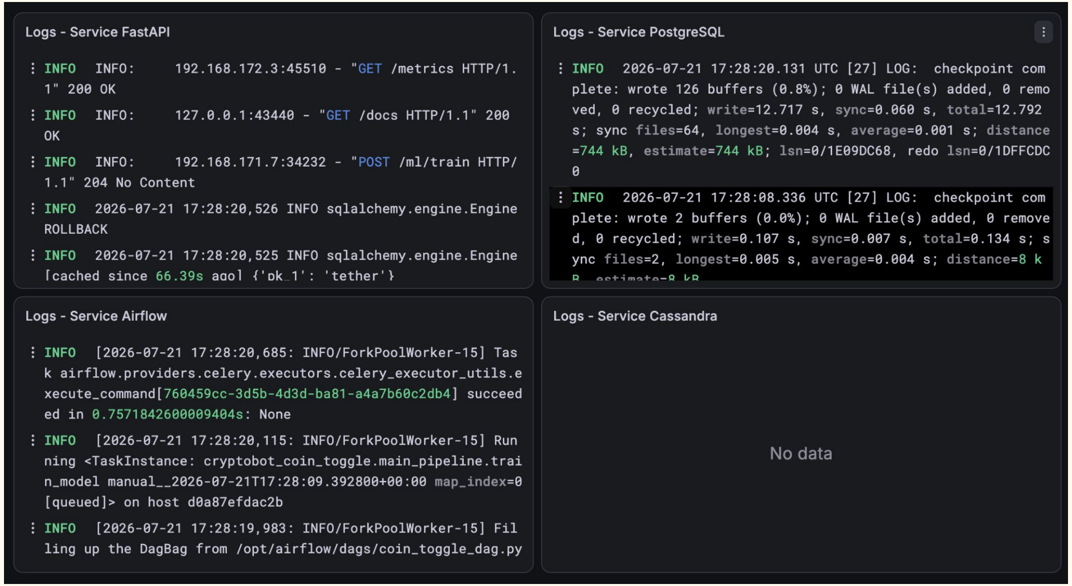
Streamlit



id	symbol	Action	Score	Consulter
bitcoin	BTC/USDT	Désactiver	0.5847	Consulter
ethereum	ETH/BTC	Désactiver	0.4992	Consulter
tether	USDT/TRY	Activer	0	
binancecoin	BNB/BTC	Activer	0	
usd-coin	USDC/BNB	Activer	0	
ripple	XRP/BTC	Activer	0	
solana	USDSOLD/USDT	Activer	0	
tron	TRX/BTC	Activer	0	
hyperliquid	HYPER/USDT	Activer	0	
dogecoin	DOGE/BNB	Désactiver	0	Consulter

Monitoring

Prometheus / Grafana



Conclusion

Conclusion

Pistes d'amélioration :

- Mise au point plus poussée de Cassandra
- Meilleurs prédictions
- Mise en place d'une vraie Landing Zone en production
- Refactoring (paramétrage de la période)
- Autres sources de données (analyse de sentiments d’articles de presse)
- Robustesse des appels à CCXT